

Uncertain Number Theory

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Abstract

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Chapter 1

Divisibility and Prime Numbers

Definition 1.1 (Divisibility). *Let $A, B \in \mathbf{U}(\mathbb{K})$. We say that A is divisible by B if for every $(a, b) \in A \times B$, a is divisible by b , and in this case, we denote $B \mid A$.*

Theorem 1.1 (Transitivity of Divisibility). *Let $A, B, C \in \mathbf{U}(\mathbb{K})$. If A is divisible by B ($B \mid A$) and B is divisible by C ($C \mid B$), then A is divisible by C ($C \mid A$).*

Proof. Assume that $B \mid A$ and $C \mid B$.

By Definition 1.1 on uncertain divisibility:

- Since $B \mid A$, for every $(a, b) \in A \times B$, we have $b \mid a$.
- Since $C \mid B$, for every $(b', c) \in B \times C$, we have $c \mid b'$.

Consider an arbitrary pair $(a, c) \in A \times C$. Since $B \in \mathbf{U}(\mathbb{K})$ is non-empty, there exists at least one element $b \in B$.

Applying classical divisibility properties in the underlying structure $\mathbb{K}$:

- From $b \in B$, we have $b \mid a$.
- From $b \in B$ and $c \in C$, we have $c \mid b$.

By the transitivity of classical divisibility ($c \mid b$ and $b \mid a \implies c \mid a$), it follows that $c \mid a$.

Since this holding holds for every pair $(a, c) \in A \times C$, by Definition 1.1, A is divisible by C , i.e., $C \mid A$.

The proof is complete. $\square$

Definition 1.2 (Prime Number). *Let $A \in \mathbf{U}(\mathbb{K})$. A is called an uncertain prime number if for every $a \in A$, a has exactly two divisors, namely a and 1.*

Definition 1.3 (Congruence). *Let $A, B, N \in \mathbf{U}(\mathbb{K})$. We say that $A \equiv B \pmod{N}$ if and only if for every $(a, b, n) \in A \times B \times N$, we have $a \equiv b \pmod{n}$.*

Proposition 1.1 (Properties of Congruence Relation). *Let $A, B, C, N \in \mathbf{U}(\mathbb{K})$. The congruence relation in the uncertain space satisfies the following properties:*

- (i) **Reflexivity:** $A \equiv A \pmod{N}$.
- (ii) **Symmetry:** If $A \equiv B \pmod{N}$, then $B \equiv A \pmod{N}$.
- (iii) **Transitivity:** If $A \equiv B \pmod{N}$ and $B \equiv C \pmod{N}$, then $A \equiv C \pmod{N}$.

Proof. The above properties follow directly from the definition of uncertain congruence relation and the corresponding congruence properties of each element $(a, b, c, n) \in A \times B \times C \times N$ in the underlying number field $\mathbb{K}$. $\square$

Proposition 1.2 (Compatibility with Addition and Multiplication). *Let $A, B, C, D, N \in \mathbf{U}(\mathbb{K})$. If $A \equiv B \pmod{N}$ and $C \equiv D \pmod{N}$, then:*

- (i) $A + C \equiv B + D \pmod{N}$,
- (ii) $A \cdot C \equiv B \cdot D \pmod{N}$.

Proof. Assume that $A \equiv B \pmod{N}$ and $C \equiv D \pmod{N}$.

According to Definition 1.3 on uncertain congruence, for every $n \in N$:

- Since $A \equiv B \pmod{N}$, for every $(a, b) \in A \times B$, we have $a \equiv b \pmod{n}$.
- Since $C \equiv D \pmod{N}$, for every $(c, d) \in C \times D$, we have $c \equiv d \pmod{n}$.

(i) Consider an arbitrary element $(a + c, b + d) \in (A + C) \times (B + D)$ and $n \in N$. From classical number theory, if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then:

$$(a + c) \equiv (b + d) \pmod{n}.$$

Since this holds for every $(a, b, c, d, n) \in A \times B \times C \times D \times N$, by Definition 1.3, we deduce:

$$A + C \equiv B + D \pmod{N}.$$

(ii) Similarly, consider an arbitrary element $(a \cdot c, b \cdot d) \in (A \cdot C) \times (B \cdot D)$ and $n \in N$. From the multiplicative compatibility property of classical congruence, we have:

$$a \cdot c \equiv b \cdot d \pmod{n}.$$

Since this holds for every $(a, b, c, d, n) \in A \times B \times C \times D \times N$, by Definition 1.3, we have:

$$A \cdot C \equiv B \cdot D \pmod{N}.$$

The proof is complete. □